Exercise 2
Aggregate Functions & Grouping

1. Select Count (id) As Total_number_of Employees
From Employees;

↳ output.

Total_number_of Employees
10

2. Select department,
Sum (Salary) As Total_salary
From employees
Group By Department
HAVING Department = 'IT';

output: If we end at Group By

department	Total_salary
IT	220 000
HR	49 000
Finance	119 000
Marketing	105 000

Note: We can end the query at Group By statement and still deduce what gets paid to employees in IT

output: If we Have Having

department	Total_salary
IT	220 000

or we can add the Having Statement after Group By & filter the aggregated data by department to get only IT

(3.) Select department,
AVG (salary) As
Average salary
From employees
Group By department
Having Department = 'HR';

→ output

department	Average salary
HR	49 000

(4.) Select Min (salary) As
lowest salary,
Max (salary) As
Higlest_salary
From employees;

→ output

lowest_salary	Higlest_salary
48 000	62 000

(5.) Select department,
Sum (salary) As
Total_salary
From employees
Group By Department;

→ output

department	Total_salary
IT	220 000
HR	99 000
finance	119 000
Marketing	105 000

(6) Select city,
Count (id) As Number_of_
employees
From employees
Group By city;

→ output

City	Number_of_employees
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

(7) Select department,
AVG (salary) As
Average_salary
From employees
Group by department
order by Average_Salary DESC;

→ output

department	Average salary
Finance	59500
IT	55 000
Marketing	52 500
HR	49 000

(8) Select department,
Sum (Salary) As total_Salary
From employees
Group by department
Having total_salary > 100 000;

→ output

department	total_salary
IT	220000
Finance	119000
Marketing	105 000

(9) Select city,
Count (id) As number-of-
employees
From employees
Group by City
HAVING number-of-employees > 1
ORDER BY Number_of_employees DESC;

→ output

City	number of employees
Chicago	3
Los Angeles	2
New York	2
San francisco	2

(10) Select department,
AVG (salary) As Average_Salary
From employees
Group by department
Order by Average_Salary DESC
Limit 1;

→ output

department	Average Salary
finance	59500